

OLIVERB MK II

A valve preamp, a passive high-pass filter, a two-track tape echo, and a spring tank — the boxes that made dub, in one plug-in.

USER MANUAL · V2.1.0 · STUDIO SERIES FACEPLATE
VST3 / AU / STANDALONE · MACOS · WINDOWS · LINUX

1 THE INTERFACE AT A GLANCE

OLIVERB is one oxblood faceplate between walnut cheeks, with six engraved modules stacked top to bottom in the order the signal flows (in the default routing). Each module is keyed by the colour of its knob caps and lamp buttons: **amber** for the Valve, **red** for the Filter, **gold** for the Echo, **lilac** for Mod, **green** for the Spring, and a black Bakelite Output knob in Master. Every module has its on/off lamp at the left (the Filter's sits at the right, beside the dial).



Figure 1 — The OLIVERB 2.1.0 faceplate at its default settings. The letters match the module table below; each module's engraved title is the heading used through this manual.

MODULE	WHAT LIVES THERE
A ● HEADER	The OLIVERB nameplate and MK II legend, the preset menu and the Bypass lamp. When a newer version is out, an UPDATE pill appears here, just left of the preset menu (see below).
B ● VALVE	Lamps Valve (on/off), At Input / At Output (where the valve sits in the chain) and Echo Amp (the same valve inside the tape machine); knobs Drive, Bias, Sag, Tone and Mix; and the 12AX7 valve at the right, which glows with how hard the stage is working.
C ● FILTER · BIG DIAL	The big red Frequency dial, its scale engraved with the chosen bank's eleven corner frequencies; Impedance, Magnetism and Character above Dynamics, Artefacts and Gain; lamps Filter , Bank A / Bank B and Pre / Post at the right.
D ● ECHO · TWO TRACK	Lamps Echo , Sync (tempo lock) and Dub Send / Held ; knobs Time, Feedback, Input, Output, Hiss, Mix and Wear. With Sync lit, Time is replaced by a note-division menu.
E ● MOD · LFO / ENVELOPE	Lamps LFO and Sync ; knobs Rate, LFO Depth, Env Depth, Sens and Speed; the Shape menu. With Sync lit, Rate becomes a division menu. Both sources drive the big dial.
F ● SPRING TANK	The Spring lamp; knobs Spring (amount), Tension (decay) and Drive.
G ● MASTER	L/R peak meters, the live signal-chain readout (it re-orders itself as you move the valve and filter), the filter spec line, and the Output knob (−24 to +12 dB).

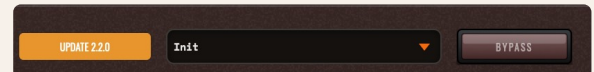
WORKING THE CONTROLS

- **Knobs** turn with a vertical or horizontal drag, or the scroll wheel. Hover, drag or scroll and the exact value appears in a bubble above the knob; the engraved scale around it shows the range at a glance.
- **Lamp buttons** light in their module's colour when on. Two-way switches name their current state on the cap: the valve's position lamp reads AT INPUT dark and AT OUTPUT lit; the filter's bank lamp reads BANK A / BANK B, its position lamp PRE / POST; the echo's send lamp DUB SEND lit and HELD dark.

RESIZABLE WINDOW

Drag the bottom-right corner (a small dotted grip appears when the mouse is over it) to scale the whole faceplate from **60 %** to **150 %** of its 980 × 930 base size. Everything is drawn as vectors, so it stays sharp at any size and on Retina / HiDPI screens. Your host's own plug-in zoom works too.

UPDATE NOTICES



While its window is open, OLIVERB checks `deestechholdings.com/updates.json` at most once a day. When a newer version exists, an amber **UPDATE x.y.z** pill appears in the header; click it for the release notes, **Download** and **Skip this version**. Nothing is sent but a plain request for the version list, and nothing runs on the audio thread.

2 INSTALLATION

MAC OS

Download `OLIVERB-macOS.zip` from the [releases page](#), unzip it, and run `OLIVERB-macOS.pkg`. It installs the VST3 and AU for all users into `/Library/Audio/Plug-Ins/VST3` and `/Library/Audio/Plug-Ins/Components`.

The installer is unsigned, so the first launch may be blocked: right-click the `.pkg` → **Open** → **Open** anyway. If the plug-in itself is quarantined:

```
xattr -dr com.apple.quarantine /Library/Audio/Plug-Ins/VST3/OLIVERB.vst3
```

Prefer a manual install? The zip also contains the raw `OLIVERB.vst3` and `OLIVERB.component` — drop them into `~/Library/Audio/Plug-Ins/VST3/` and `~/Library/Audio/Plug-Ins/Components/`.

WINDOWS

Download `OLIVERB-Windows.zip`, unzip, and run **OLIVERB-Windows-Setup.exe**. It installs to `C:\Program Files\Common Files\VST3\OLIVERB.vst3`. SmartScreen will warn about an unrecognised app: click **More info** → **Run anyway**. For a manual install, copy the `OLIVERB.vst3` folder into `C:\Program Files\Common Files\VST3\`.

LINUX

Download `OLIVERB-Linux.zip` and copy `OLIVERB.vst3` to `~/.vst3/`.

After installing, rescan plug-ins in your DAW. OLIVERB appears under **Deestech** in the Fx / Filter / Delay / Reverb categories. **Upgrading** from v1 or v2.0: install over the top. 2.1.0 changes only the faceplate — the sound and every parameter are the same, so sessions reload exactly as saved (sessions from v1 load with the valve at its default settings).

3 FIRST SOUND IN SIXTY SECONDS

1. Insert OLIVERB on a drum loop or a full mix.
2. Open the preset menu in the header and pick **Waterhouse Rockers** — the classic chain: valve into filter into echo into spring.
3. Turn the valve **Drive** up and watch the 12AX7: it glows brighter, and the sound thickens and compresses but doesn't get louder. That's the tube.
4. Grab the big **Frequency** dial and sweep it up and back down. The corner peak, the switch clacks, the way the echo tail follows the dial — that's the plug-in.
5. Turn **Feedback** past 80 % and tap **Dub Send** so it goes dark and reads **Held**: the input stops feeding the machine but the loop keeps regenerating. That's a dub-out.
6. Tap it back to **Dub Send** and carry on.

4 SIGNAL FLOW

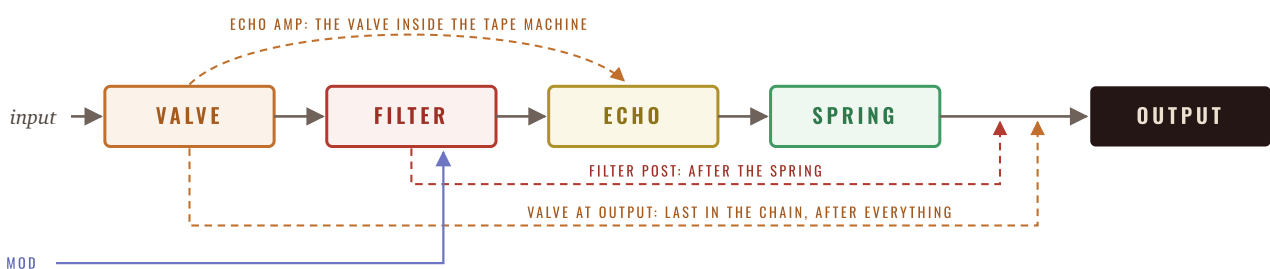


Figure 2 — The signal chain. The valve sits first by default, or last with **At Output** lit; **Echo Amp** puts the same valve inside the tape machine. The **Mod** module steers the filter's frequency dial wherever the filter sits.

The **Valve** colours everything after it at the front, or becomes a master valve stage across the whole mix, echo tails and spring included, at the back. The filter's **Pre** / **Post** lamp is the other big routing decision:

- **Pre** — the filter feeds the echo, so every repeat inherits the dial position, and sweeping the dial drags the echo tail with it. This is the performance position.
- **Post** — the filter sits across the finished signal, echo tails included. Use it to carve the whole wet mix at once.



You never have to remember the routing: the **chain readout** in the Master module spells out the current order — for example **IN > VALVE > FILTER > ECHO > SPRING > OUT** — and re-orders itself as you flip *At Output* or *Post*. A switched-off valve drops out of it.

Everything runs at 2× oversampling internally; the extra latency is reported to your host automatically, so tracks stay aligned.

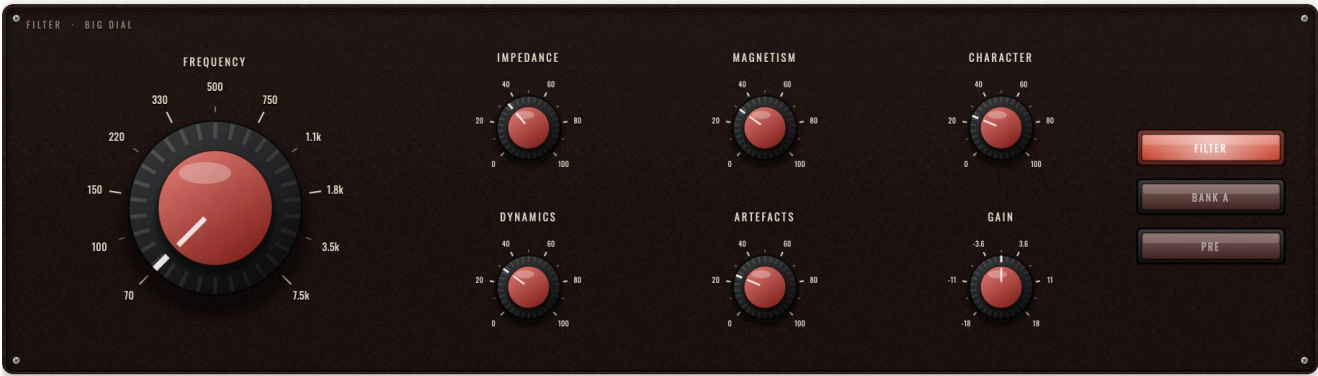
5 THE VALVE — PREAMP



A single-ended triode line amplifier with an output transformer. Its transfer curve is asymmetric — grid conduction flattens the positive swing quickly, cut-off softens the negative swing slowly — which is why it makes even harmonics (warmth) rather than the odd-only fuzz of a symmetric clipper. The drawn **12AX7** at the right of the module glows with how hard the valve is actually working.

CONTROL	RANGE (DEFAULT)	WHAT IT DOES
Valve	(On)	Switches the valve stage in or out.
Drive	0–100 % (35 %)	Gain into the grid, up to +36 dB. Level is compensated on the way out, so Drive changes the <i>shape</i> of the sound far more than its loudness. Zero is a clean line amp (about 1 % second harmonic); full is a valve on its knees.
Bias	0–100 % (50 %)	The operating point. Cold is near-symmetric and stays clean until pushed; hot is asymmetric from the first volt and thick at any level — the even-harmonic knob.
Sag	0–100 % (30 %)	The supply giving way under load: gain drops and bias goes colder as the signal rises. Compression, bloom, a stage that pushes back.
Tone	0–100 % (40 %)	The output transformer: iron bump below 100 Hz, loss at the very top (19 kHz down to 6.5 kHz), and the core rounding off what's still too big for it.
Mix	0–100 % (100 %)	Parallel blend against the dry signal.
At Input / At Output	(At Input)	<i>At Input</i> (dark) puts the valve first in the chain; <i>At Output</i> (lit) puts it last, after the spring.
Echo Amp	(On)	Hands the same valve curve to the tape machine's record amplifier, scaled by Drive. Off restores the original solid-state record amp.

6 THE FILTER — BIG DIAL



A passive, inductor-based high-pass, third order (18 dB/octave). No active parts — its character comes from termination and core saturation, not resonance. The dial's engraved scale always shows the corner frequencies of the bank you have selected, so it re-labels itself when you switch Bank A / Bank B.

CONTROL	RANGE (DEFAULT)	WHAT IT DOES
Filter	(On)	Switches the filter in or out.
Frequency	1–11 (1)	The big dial: eleven switch positions — see the table below.
Bank A / Bank B	A / B (A)	Capacitor bank. A is the classic broad sweep; B sits lower and tighter.
Impedance	0–100 % (35 %)	Termination. At 0 % the network is properly terminated and flat; open it up and the corner lifts by several dB. This peak is the sound of the hardware.
Magnetism	0–100 % (30 %)	How hard the inductor cores saturate. Signal level pushes the corner upward and generates harmonics.
Character	0–100 % (25 %)	Non-linearity in the damping path — harmonic content, independent of the core.
Dynamics	0–100 % (30 %)	How fast the core tracks the signal. Slow = a gentle breathing lift; fast = it behaves like an envelope filter.
Artefacts	0–100 % (25 %)	Switch thump. At 0 % the corner glides between positions (the hardware can't). At 100 % it clacks like the real rotary.
Gain	±18 dB (0)	Internal trim after the filter.
Pre / Post	(Pre)	Filter before the echo, or across the finished signal — see §4.

SWITCH POSITIONS

STEP	1	2	3	4	5	6	7	8	9	10	11
Bank A	70	100	150	220	330	500	750	1.1k	1.8k	3.5k	7.5k
Bank B	50	80	120	170	240	350	520	800	1.3k	2.4k	5.0k

All values in Hz — exactly as engraved around the dial.

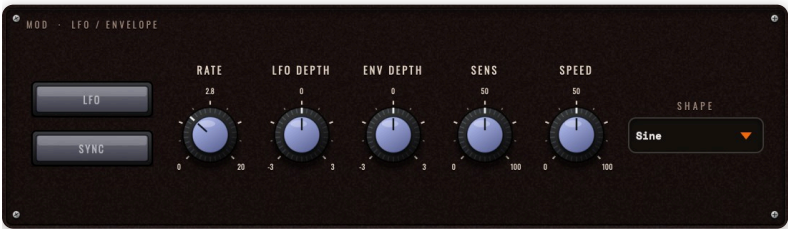
7 THE ECHO — TWO TRACK



A studio tape recorder used as an echo: record head → tape → replay head, output patched back to the input. Each pass through the loop loses a little top end and gains a little saturation, so long tails get darker and thicker rather than louder. With the valve's Echo Amp lit, that saturation is the triode curve.

CONTROL	RANGE (DEFAULT)	WHAT IT DOES
Echo	(On)	Switches the echo in or out.
Time	20–2000 ms (375)	Tape delay. Changing it drags the transport, so pitch bends on the way — as tape does.
Sync	(Free)	Lock the delay to the host tempo. When lit, the Time knob is replaced by a note-division menu, 1/16 to 1/1 with triplet (T) and dotted (.) values (default 1/4).
Feedback	0–100 % (34 %)	Loop regeneration. Unity sits at 80 % — above that the machine self-oscillates on purpose, limiting into the record amp instead of exploding.
Dub Send / Held	(Dub Send)	<i>Dub Send</i> (lit) feeds the input into the machine. <i>Held</i> (dark) closes the door: the input stops, but whatever is on the loop keeps circulating.
Input	0–100 % (57 %)	Level into the record amp. Push it to saturate the tape harder.
Output	0–100 % (60 %)	Level out of the machine.
Hiss	0–100 % (18 %)	Tape noise, recorded <i>to</i> the tape — it recirculates and builds with feedback instead of sitting on top.
Mix	0–100 % (32 %)	Dry/wet balance for the echo.
Wear	0–100 % (45 %)	Machine condition: one knob scaling flutter, HF loss, head bump and record-amp drive from freshly aligned to tired.

8 MOD — LFO / ENVELOPE



Both modulation sources push the **Frequency** dial, in octaves relative to the switch position, so everything the filter does — corner peak, core saturation, character — rides along with the sweep. Modulation is shared by both channels, so the stereo image holds.

LFO

CONTROL	RANGE (DEFAULT)	WHAT IT DOES
LFO	(Off)	Enables the LFO.
Shape	5 shapes (Sine)	Sine, Triangle, Saw Down, Square, S+H — the menu at the right of the module. Square and S+H are lightly smoothed: the corner snaps, the zipper doesn't.
Rate	0.02–20 Hz (0.8)	Free-running speed.
Sync	(Free)	Lock the rate to the host tempo. When lit, the Rate knob becomes a division menu, 1/16 to 1/1 (default 1/2); synced sweeps re-align to the bar.
LFO Depth	±3 oct (0)	How far the dial travels. Negative inverts the sweep.

ENVELOPE FOLLOWER

Follows the input signal (pre-filter), so it responds to what you play, not to what the filter is already doing. It is always available — Env Depth at 0 is off.

CONTROL	RANGE (DEFAULT)	WHAT IT DOES
Env Depth	±3 oct (0)	Bipolar: positive opens the filter on hits (auto-wah); negative ducks it out of the way.
Sens	0–100 % (50 %)	How hard the follower listens.
Speed	0–100 % (50 %)	How fast it moves.

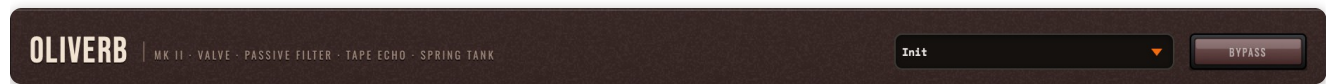
9 THE SPRING — TANK



A two-tank spring reverb. Springs are dispersive — highs travel through the coil faster than lows — so a transient smears into the characteristic descending *boing* that room reverbs never produce.

CONTROL	RANGE (DEFAULT)	WHAT IT DOES
Spring (lamp)	(On)	Switches the tank in or out.
Spring	0–100 % (22 %)	How much of the tank is in the output.
Tension	0–100 % (55 %)	Decay time of the tank.
Drive	0–100 % (25 %)	Level into the send transducer. This is where the crash lives — hit it hard and transients splash.

10 MASTER & GLOBAL



CONTROL	RANGE (DEFAULT)	WHAT IT DOES
Output	-24 to +12 dB (0)	Final level trim — the black knob at the right of the Master module.
Bypass	(Off)	True bypass of the whole plug-in. The lamp in the header lights red when bypassed.
Presets	14 factory	The menu in the header. Loading a preset starts from the defaults and applies its own settings (§11).
Meters	L / R peak	Twenty-segment peak ladders covering the top 48 dB of the final output, after the Output trim.
Chain readout	—	The live signal order (§4), with the filter's slope and oversampling beneath it.

Each of the four processing modules also has its own on switch, so OLIVERB can serve as just a valve, just a filter, just an echo, or just a spring.

11 FACTORY PRESETS

PRESET	WHAT IT SHOWS
Init	Everything at defaults — the valve on at a gentle 35 %.
Waterhouse Rockers	The classic chain at working settings, valve warm — the place to start.
Valve Warmth	Valve only, everything else off: a tube line amp on a bus or a mix.
Hot Preamp	Valve only, driven hard with heavy sag: a mic pre on its knees.
Snare Throw	High feedback, hot input: hit it with one snare and ride the tail.
Big Dial Sweep	Open termination, heavy magnetism, audible switch clacks — for performing the dial.
Tape Wash	Long, worn, hissy repeats that melt into the spring, valve tone thick.
Glowing Tape	The valve <i>at output</i> , driven, with a hot tape loop — everything cooks together.
Held Echo (Dub Out)	Send is <i>Held</i> and feedback at unity: an infinite loop, filter across the output. Light Dub Send to let new signal in.
Spring Crash	The tank up front, driven hard.
Roots Bass Tighten	Filter and valve only, step 1, a little core saturation — a bass-tightening tool, no wet signal at all.
Auto Wah Skank	Envelope follower opening the dial on hits. Guitars and clavs.
Tidal Sweep	Tempo-synced triangle LFO sweeping two octaves over a whole bar.
Siren	Everything at maximum, valve included. You were warned.

12 TECHNIQUES

Valve as a bus tool. Load *Valve Warmth*. Drive 30–50 %, Bias 60 %, Sag 20 %: glue and weight without obvious distortion. The 12AX7 should flicker, not burn.

Pushed preamp. Drive past 70 % with Sag up: the stage compresses into itself and blooms after transients. Bias hot for thickness, cold for a cleaner crunch.

Valve at the back. Light *At Output* so the valve sits after the spring: echo tails and splashes get rounded off by the same tube, and hot feedback stops sounding digital.

The dub throw. Feedback high, Mix high, *Dub Send* lit. Un-mute one hit — a snare, a vocal word — then tap the lamp to *Held*. The hit circulates and decays on its own while the dry signal carries on. Tap back to *Dub Send* for the next throw.

Ride the dial. With the filter on *Pre*, sweep Frequency during the echo tail: every repeat re-inherits the new corner and the whole tail bends. Add Artefacts for the hardware clack on each step.

Pitch-bend echoes. Automate Time. The transport drags, so repeats bend like varispeed tape. Small moves = subtle warble; big jumps = dive-bombs.

Self-oscillation instrument. Feedback past 80 %, no input: the machine sings on its own. Tune it with Time, filter it with the dial, crash it into the spring.

Auto-wah. Env Depth positive, filter step 4–6, Impedance up. Sens sets how hard it listens, Speed how fast the corner chases the playing.

Reverse duck. Env Depth negative: the filter closes on hits and blooms back open in the gaps — an ungate for pads and textures.

13 TROUBLESHOOTING

SYMPTOM	FIX
Plug-in doesn't appear in the DAW	Rescan plug-ins. Confirm the <code>.vst3</code> is in the system VST3 folder (§2). On Apple-silicon Macs check the DAW isn't running in Rosetta with an arm64-only scan cache.
macOS blocks the installer or plug-in	Right-click → Open on the <code>.pkg</code> , or clear quarantine with the <code>xattr</code> command in §2.
Windows SmartScreen warning	More info → Run anyway. The installer is unsigned, not unsafe.
Window too big or too small	Drag the bottom-right corner: the faceplate scales from 60 % to 150 %.
An UPDATE pill appeared	A newer OLIVERB is out. Click it for the notes and download, or Skip this version to hide it until the next release.
Sound is late or flammng	OLIVERB reports its 2× oversampling latency to the host; enable your DAW's plug-in delay compensation.
Everything sounds slightly thicker than v1	That's the valve at its default 35 %. Switch the Valve lamp off, or Drive to 0, for the v1 sound.
Echo tail never dies	Feedback is at or above 80 % (unity). That's a feature — pull it down, or light <i>Dub Send</i> with no input.
Output slammed after big dial moves	Open Impedance plus hot echo feedback genuinely adds level. Use the filter Gain or the Master Output knob. The output stage is hard-limited, so it cannot run away.

14 SPECIFICATIONS

- **Valve:** single-ended triode model, unity-normalised asymmetric transfer, +36 dB drive range, level-compensated, supply sag, transformer voicing
- **Filter:** third-order (18 dB/oct) passive constant-k model, TPT state-variable core, stable under audio-rate modulation
- **Echo:** 20–2000 ms, wow & flutter, per-pass HF loss, in-loop record-amp saturation (solid-state or valve)
- **Spring:** two dispersive tanks (33.7 / 41.9 ms coil transit), twelve allpass sections each
- **Processing:** 32-bit float, 2× oversampled (half-band polyphase IIR), latency-compensated
- **Interface:** Deestech Studio Series faceplate, vector-drawn; 980 × 930 base size, resizable 60–150 %; optional update notice
- **Formats:** VST3, Audio Unit (macOS), Standalone application
- **Platforms:** macOS 10.13+ (universal: Apple silicon + Intel), Windows 10+ (64-bit), Linux
- **Source & license:** MIT, at github.com/damielolar-CODE/oliverb

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